

# **INTERSECTION OF HUMAN TRAFFICKING AND DRUG TRAFFICKING: IDENTIFYING PATTERNS OF THE USAGE OF ILLICIT SUBSTANCES AS A MEANS OF CONTROL DURING HUMAN TRAFFICKING**

by

**ARISHA SHYPEL**

Dissertation submitted in partial fulfilment of the requirements for the degree of BSc in Security and Crime Science (UCL) of the University of London in 2024.

UNIVERSITY COLLEGE LONDON

UCL DEPARTMENT OF SECURITY AND CRIME SCIENCE

This Dissertation is an unrevised examination copy for consultation only and it should not be quoted or cited without the permission of the Chairman of the Board of Examiners of the BSc in Security and Crime Science (UCL)

**BSc Dissertation**  
**University College London**  
**Department of Security and Crime Science**  
**BSc Dissertation**

**STUDENT'S DECLARATION**

I, Arisha Shtypel, hereby declare that this dissertation is my own original work and that I have clearly identified and acknowledged all source material used. No part of this dissertation contains material previously submitted to the examiners of this or any other university, or any material previously submitted for any other examination.

This dissertation is 8876 words in length (including abstract but excluding reference list and reasonable use of tables and figures).

Student's signature: Arisha Shtypel

Date: 02.09.2024

**University College London  
Department of Security and Crime Science**

**BSc Dissertation**

**SUPERVISOR'S DECLARATION**

Name of student: Arisha Shtypel

Name of (primary) supervisor: Dr Kevin Chetty

I confirm that the student named above has undertaken this dissertation under my supervision, attended meetings with me as requested and provided me with adequate information about the progress of the research.

Supervisor's signature: .....  .....

Date: 21/08/2024



**Student name: Arisha Shtypel**

**Supervisor name: Kevin Chetty**

**Title of proposed project: Exploring the Relationship Between Human Trafficking and Controlled Substances**

**Project ID: 728**

Upon review of the materials that you provided, the Department of Security and Crime Science Ethics Committee has decided that your proposed research is exempt from requiring approval by the UCL Research Ethics Committee. This is because the proposed research either:

- does not involve human participants and/or does not involve the collection and/or use of data derived from human individuals, or
- corresponds to one or more of the following UCL exemption criteria (<https://ethics.grad.ucl.ac.uk/exemptions.php>):

| Ethics exemption |                                                                                                                                                                                                                                                                                                                               |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.               | Research involving information freely available in the public domain. For example, published biographies, newspaper accounts of an individual's activities and published minutes of a meeting, whilst still personal data under the Data Protection Act would not require ethics review.                                      |
| 2.               | Research involving anonymised records and data sets that exist in the public domain. For example, datasets available through the Office for National Statistics or the UK Data Archive where appropriate permissions have already been obtained and it is not possible to identify individuals from the information provided. |
| 3.               | Studies of public behaviour that are purely observational (non-invasive and non-interactive), unless the recorded observations identify individuals (names, photographs) which could place them at risk of harm, stigma or prosecution.                                                                                       |
| 4.               | Research involving the use of non-sensitive, completely anonymous educational tests, survey and interview procedures when the participants are not defined as "vulnerable" and participation will not induce undue psychological stress or anxiety.                                                                           |
| 5.               | Research involving the use of educational tests, survey and interview procedures on human participants in the public arena (e.g. elected or appointed public officials, candidates for public office, artists).                                                                                                               |

Should your project substantially change from what you have proposed, you will need to go through the ethics process again.

**Signed:**

**Dated: 9/11/2023**

**Dr Kartikeya Tripathi**

Chair, Departmental Ethics Committee  
Department of Security and Crime Science  
University College London

University College London, Gower Street, London WC1E 6BT  
Tel: +44 (0)20 7679 2000  
email@ucl.ac.uk  
www.ucl.ac.uk

## **ACKNOWLEDGEMENTS**

The study is a product of painstaking work on a fascinating and important topic. The project would not have taken its final form without the guidance on every step of the way from my supervisor Dr Kevin Chetty. I am especially grateful to him for helping me find and choose the most versatile data possible in the context of Human Trafficking and formulate and narrow down the research area, when I was struggling to come up with ideas that could be realized with limited data availability. I also would like to thank Dr Kevin Chetty for supporting me through the periods when I thought I have hit dead ends with the analysis and providing his expertise on finding solutions.

## **ABSTRACT**

Human Trafficking (HT) and Drug Trafficking (DT) are highly pervasive and dangerous illicit markets. The two markets are interconnected and have an intricate interdependent relationship. However, literature does not often explore their interconnectedness in further detail. Most research focuses on either HT or DT. After closer inspection it is evident that the points of connection within HT and DT spawn through different dimensions of interactions, such as the economic, social, and individual. It would be beyond the scope of the present research to attempt to provide a complete account of all the points of intersection at hand. Thus, a specific point of intersection was chosen for exploration. One manifestation of the intersection of HT and DT is the usage of drugs on the victims of HT.

This study will attempt to explore the intersection between HT and DT through the analysis of HT cases where control over victims was established by the means of the usage of illicit substances. The present study's main purpose is to identify patterns of the usage of illicit substances on HT victims and attempt to explain the reasons behind their formation. The data used for the present study comes from the Counter Trafficking Data Collaborative (CTDC), which is a global hub of HT cases recorded by various organisations around the world.

Principle Component Analysis (PCA), a Machine Learning technique, was used to analyse the cases within the CTDC dataset which indicated that control over the victim was established through illicit substances. The main idea was to reduce the dimensionality of the large CTDC dataset and by doing so uncover the most prominent factors that contributed to cluster formation. Thus, allowing for interpretation of why these specific factors play such a significant role. PCA revealed that the most prominent factors are demographic: gender and age, geographical: citizenship and country of exploitation and the presence or absence of other means of control.

# TABLE OF CONTENTS

**INTRODUCTION ..... 1**

**LITERATURE REVIEW ..... 7**

2.1 Globalisation: Economics of the Intersection of Human Trafficking and Drug Trafficking.....8

2.2 Theoretical Background: Coercion and Methods of Control ..... 10

2.3 Human Trafficking: Geographical Patterns..... 14

**METHODOLOGY ..... 16**

3.1 Data..... 16

3.2 Data Limitations ..... 17

3.3 Pre-Processing ..... 18

3.4 Explorative Data Analysis..... 23

**RESULTS..... 24**

4.1 PCA: Explained Variance..... 24

4.2 PCA: Principal Components Composition ..... 25

4.3 PCA: Clusters ..... 28

**DISCUSSION ..... 31**

5.1 Key Findings ..... 32

5.2 Limitations..... 33

5.3 Implications ..... 34

**CONCLUSION ..... 35**

**BIBLIOGRAPHY ..... 36**

## List of Figures

- Figure 1      Total victims detected per 100,000 population, 2003-2020 (Source: UNODC Global Report on Human Trafficking 2022)
- Figure 2      Detected victims of trafficking by form of exploitation, 2020 (or most recent), 2020 (Source: UNODC Global Report on Human Trafficking 2022)
- Figure 3      Trend: Detected victims of trafficking, by age group and sex 2004-2020 (Source: UNODC Global Report on Human Trafficking 2022)
- Figure 4      Power and Control Wheel
- Figure 5      Kernel density estimates for the marginal distributions of the observed data (blue) and the  $m=3$  densities per variable calculated from the imputed data (thin red lines)
- Figure 6      PCA Explained Variance
- Figure 7      PCA 3D Visualisation (from different angles)

## List of Tables

- Table 1      Variable Classes and Corresponding Imputation Methods
- Table 2      Results Summary

## Appendixes



Appendix I     Variable Explanation Table

Appendix II    Countries Grouped by Sub-Regions

### **Acronyms**

|       |                                          |
|-------|------------------------------------------|
| CTDC  | Counter Trafficking Data Collaborative   |
| DT    | Drug Trafficking                         |
| EU    | European Union                           |
| HT    | Human Trafficking                        |
| MICE  | Multiple Imputation by Chained Equations |
| OC    | Organised Crime                          |
| RCT   | Rational Choice Theory                   |
| RQ    | Research Question                        |
| UN    | United Nations                           |
| UNODC | United Nations Office on Drugs and Crime |

# INTRODUCTION

## Chapter I

Slavery is a dark stain on the pages of world history. Its roots trace back to the classical world, appearing in the Babylonian laws of Eshnunna and the code of Hammurabi (Black, 2011), with its apogee being the transatlantic slave trade, which coincided with the dawn of modern Europe in the 15th century and the beginning of the 16<sup>th</sup> century (Rawley and Behrendt, 2005). Although since then, with advancements in human rights, slavery and the slave trade have been abolished, more subtle means of human trade have emerged, fuelled by economic and social inequalities. In other words, the chains of slavery have not been broken but merely transformed and hidden, manifesting in the form of HT.

As defined by the United Nations Office on Drugs and Crime (UNODC):

“Human Trafficking is the recruitment, transportation, transfer, harbouring or receipt of people through force, fraud or deception, with the aim of exploiting them for profit. Men, women and children of all ages and from all backgrounds can become victims of this crime, which occurs in every region of the world. The traffickers often use violence or fraudulent employment agencies and fake promises of education and job opportunities to trick and coerce their victims”

*United Nations: Office on Drugs and Crime. Human-Trafficking.*

In 2021, the Global Initiative against Transnational Organised Crime highlighted that "human trafficking [was] the most pervasive of all criminal markets globally" (The Global Organised Crime Index, 2021), and even though financial crime emerged as the most pervasive criminal market in 2023, HT was a close second (The Global Organised Crime Index, 2023). UNODC

provides regular reports on HT to indicate current and emerging trends. Figures 1 to 3 were taken from Global Report on Human Trafficking 2022. They provide an overview of the larger trends within HT. It is evident from Figure 1 that throughout the years HT has been increasing at a rather steady rate. Although there is a drop in 2020 compared to 2019, the rate of HT is still alarmingly high.

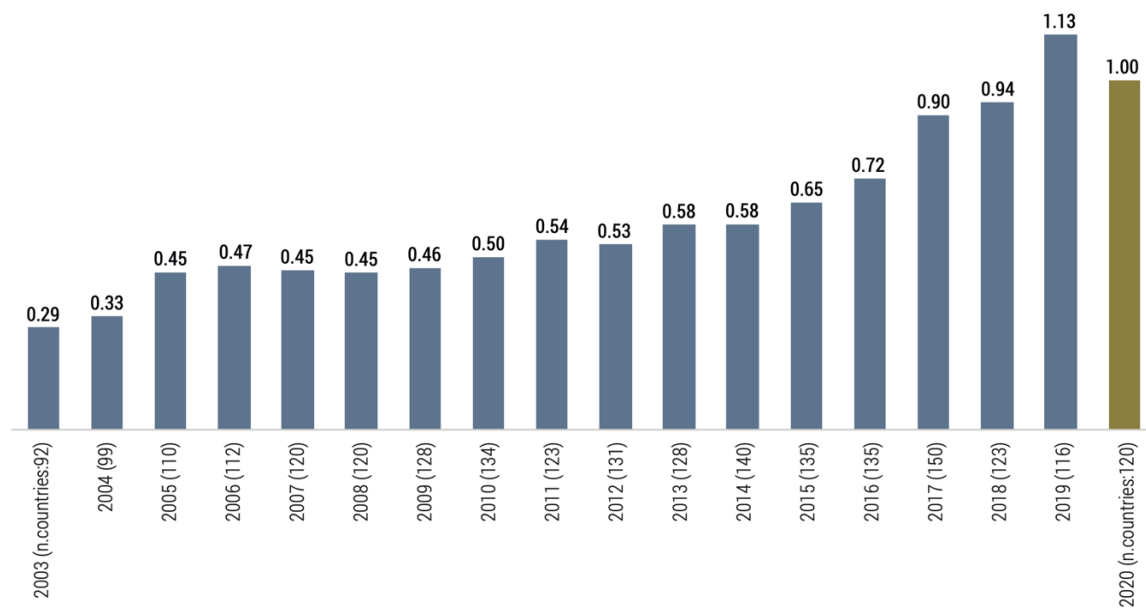


Figure 1: Total victims detected per 100,000 population, 2003-2020 (Source: UNODC Global Report on Human Trafficking 2022)

To further analyse HT, it is important to understand the different types of exploitation and their distribution. Figure 2 indicates that trafficking for sexual exploitation and forced labour are by far the most prominent types.

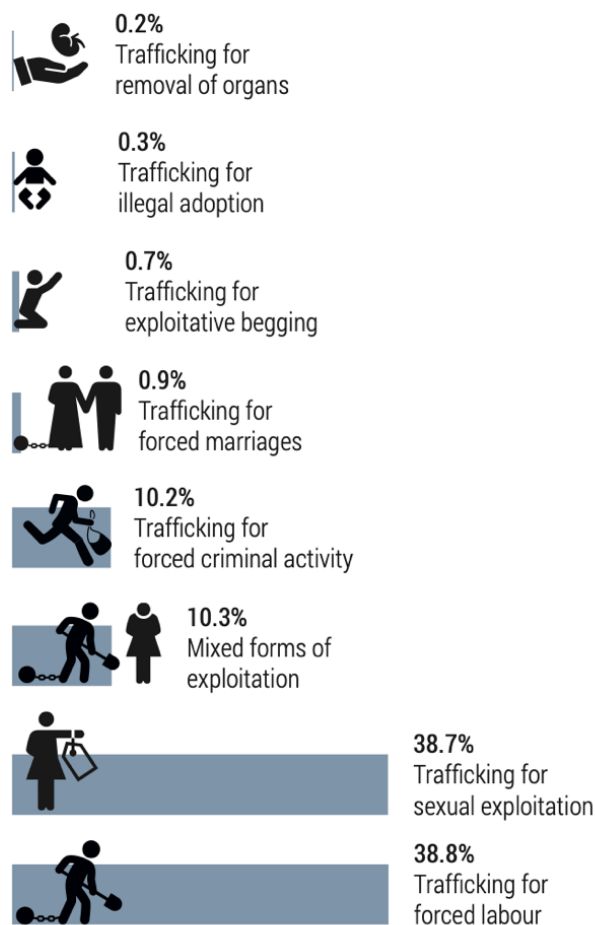


Figure 2: Detected victims of trafficking by form of exploitation, 2020 (or most recent), 2020 (Source: UNODC Global Report on Human Trafficking 2022)

Throughout the years women have remained the largest demographic group within HT (see Figure 3). Although it is also evident from the graph that throughout the years the percentage of women compared to men, boys and girls is gradually decreasing. Interestingly, the percentage of men, girls and boys all show an opposite trend and are increasing. However, the presented graphs and statistics do need to be taken critically as the Dark Figure of Crime remains and a large portion of HT cases are unreported and unrecorded.

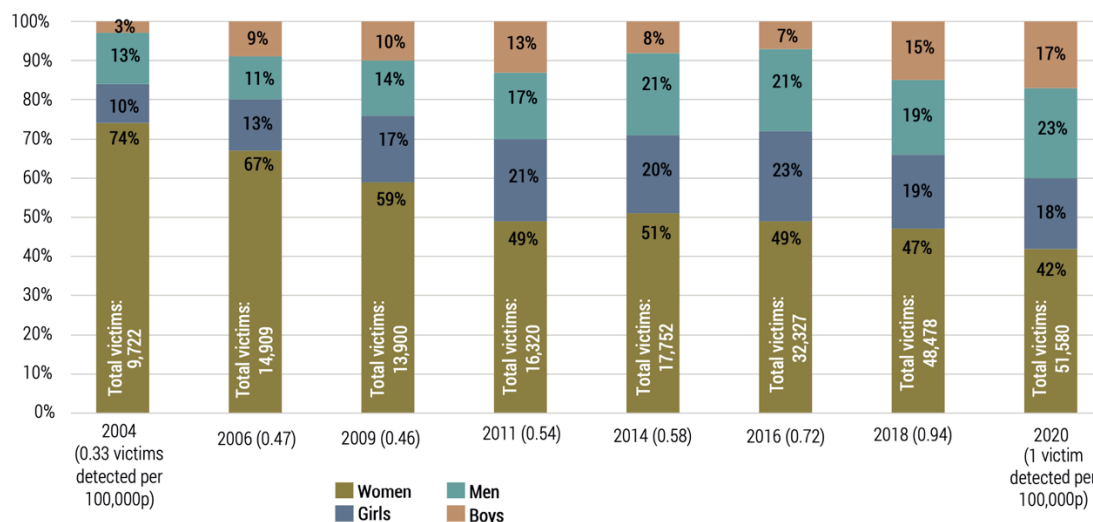


Figure 3: Trend: Detected victims of trafficking, by age group and sex 2004-2020 (Source: UNODC Global Report on Human Trafficking 2022)

Many stakeholders are actively addressing the pervasive issue of HT. The UN adopted the Palermo Protocol in 2000 to identify and aid victims from various demographic groups and to detect all forms of HT (Protocol to Prevent, Suppress, and Punish Trafficking in Persons, Especially Women and Children, Supplementing the United Nations Convention Against Transnational Organized Crime, 2000). In addition, other individual states and unions work to prevent HT in their territories. In the United Kingdom, the Modern Slavery Act 2015 outlines the offences that fall under the modern slavery umbrella. It also lists the prevention and protection mechanisms in place. To combat HT, the European Union (EU) and the United Nations (UN) run numerous campaigns, such as "End Human Trafficking" and the "Blue Heart Campaign." However, despite reflecting the complexity of the crime on paper, accounting for it in practice can be challenging. Campaigns and non-government organisations' activities frequently focus on specific types of victims or certain types of trafficking. For instance, ECPAT International dedicates its efforts to assisting children trafficked for sexual exploitation. Inadvertently, this effort disqualifies victims who do not meet the specific criteria for aid. Furthermore, as the scope of HT becomes clear, organizations gather data, and

campaigns and the media shape public perception and awareness. Thus, translating a specific victim image can be harmful, as the process of selecting a certain type of victim for an anti-trafficking campaign is inherently exclusive and fortifies misrepresentation (O'Brien, 2013).

One of the reasons for this is the frequent presentation of HT victims in Nils Christie's paradigm of the "ideal victim" (Christie, 1986). However, this type of portrayal can be misleading as it often omits the less commonly observed categories of victims (Wilson and O'Brien, 2016). In the case of HT, it is crucial to closely monitor each victim profile within the identified victims, as it is not possible to empirically examine the proportion of hidden cases. In this context, it appears that victimology is a highly useful tool. Due to the hidden nature of HT, a lot of the cases go unnoticed, and it can be difficult to explain the tendencies numerically. Victimology aids in identifying hidden patterns. The discovery of such patterns is crucial to providing necessary support to identified victims as well as reducing victimisation.

HT has received a lot of scholarly attention, particularly in recent years. Scholars have examined the patterns from various perspectives: broad, such as geography, or specific, such as the type of work in HT for sexual exploitation. Another interesting area to explore is connection to other crimes (Decker, 2015). While HT has also been explored in the contexts of adjacent crimes to some extent, the analysis of existing literature revealed a lack of knowledge about the interdependency between HT and DT.

As defined by UNODC:

“Drug trafficking is a global illicit trade involving the cultivation, manufacture, distribution and sale of substances which are subject to drug prohibition laws.”

*United Nations: Office on Drugs and Crime. Drug Trafficking.*

It is known that victims of HT are sometimes used as drug couriers (Shelley, 2012). We also know that certain cases involve the use of drugs to exert control over a HT victim. Interestingly, the majority of existing literature focuses on the aftermath of drug exposure during HT. Health policy and systems, social work, and mental health compose the majority of this literature (Sweileh, 2018). However, little is known about the victimological perspective on this issue, or, more specifically, victim profiles.

Despite the numerous intersection points, the literature has not depicted the relationship between HT and DT in this context. This dissertation will aim to contribute to this area of knowledge by analysing the victim profiles and patterns in the use of illicit substances as a means of control during HT. The victimological approach was chosen as it proved successful in analysing HT in general. However, no such research has been done from the perspective of the usage of drugs on the victims of HT. Establishing the patterns could prove useful in 1) identifying and assisting HT victims in emergency rooms. 2) Treating identified HT victims 3) Developing more informed HT policies. 4) Carrying out more multifaceted campaigns.

The following Research Question has been put forward:

**Are there patterns within the usage of illicit substances as a means of control during human trafficking?**

*Hypothesis<sub>0</sub>*: There are no distinct patterns of the usage of illicit substances as a means of control during human trafficking

*Hypothesis<sub>1</sub>*: There are distinct geographical patterns of the usage of illicit substances as a means of control during human trafficking

*Hypothesis<sub>2</sub>*: There are distinct demographic patterns of the usage of illicit substances as a means of control during human trafficking

*Hypothesis<sub>3</sub>*: There are distinct patterns of the usage of illicit substances as a means of control during human trafficking based on the type of exploitation the victim endures

The paper is structured as follow. In Chapter II, a thematic literature review is provided to theoretically inform the present study. The thematic sections of the literature review are 1) Globalisation: Economics of the Intersection of Human Trafficking and Drug Trafficking 2) Theoretical Background: Coercion and Methods of Control and 3) Human Trafficking: Geographical Patterns. Chapter III presents the methodology for the present research study. Chapter IV illustrates the results from Principal Component Analysis. Chapter V discusses the results in broader contexts, highlights associated limitations and provides recommendations for policy and research. Chapter VI concludes the dissertation.

## **LITERATURE REVIEW**

### **Chapter II**

This chapter delves into the existing literature that served as the foundation for this dissertation. The literature has been divided into three main sections thematically. Each section aids in placing the study within the relevant theories and research while highlighting relevant gaps. To conceptualise the thematic sections, the dissertation's research question was dissected into three areas of interest. The first section will look at the literature, which emphasises the intersection of HT and DT. Particular emphasis will be placed on economic aspects and



globalisation. It will aim to answer the questions of how and why HT and DT intersect. The second section will review existing theories and frameworks that are used to understand the nature of control and coercion. It will lay the groundwork for understanding various forms of control, as well as why they are an essential component of HT. The third section will focus on understanding HT patterns and the rationale behind them. A comprehensive analysis of existing HT patterns should provide greater context into the patterns of the usage of illicit substances on victims of HT.

## **2.1 Globalisation: Economics of the Intersection of Human Trafficking and Drug Trafficking**

HT and DT are both manifestations of Organized Crime (OC). To create a solid foundation for analysis and interpretations of their intersections it is important to look at OC first. Interpol proposed the following definition:

“Organized criminal networks are involved in many different types of criminal activities spanning several countries. These activities may include trafficking in people, drugs, illicit goods and weapons, armed robbery, counterfeiting and money laundering. With revenues estimated in the billions, their criminal enterprises closely resemble those of legitimate international businesses. They have operating models, long-term strategies, hierarchies, and even strategic alliances, all serving the same purpose: to generate the most profits with the least amount of risk.”

*Interpol, Organized Crime*

Similar to legitimate businesses, globalisation has been highly beneficial for OC networks. In concise terms, globalisation can be defined as “the world-spanning intensification of interconnectedness” (Vertovec 2009, 54). Globalisation has primarily facilitated OC networks through two vectors. Firstly, the decline in regulations and increased freedom allowed OC

networks to expand on the global market (Aronowitz, 2009). Rising demand for illegal goods consolidated their influence. Secondly, due to the deconstruction of the centralised state and administrative decentralisation, OC networks strengthened their positions as local players (Armao, 2014).

However, globalisation also brought an increase in competition. Competitiveness is one of the characteristics that is particularly prevalent in OC groups. OC is not merely a “crime that is organised”, it has its roots in the idea of power and control and the exclusivity of power and control (Schelling, 1984). Consequently, OC seeks to establish a monopoly, exemplified by its distinct monopolistic characteristics. Characteristics unique to organised crime groups, such as invisibility, dynamic nature, profitability, and social irresponsibility, explain the higher degree of competitiveness and successful monopolisation (Armao, 2014). Organised crime groups adopt a two-fold strategy of entrenchment (territorial conquest) and expansion (projection of power), producing OC clusters to achieve and sustain control (Armao, 2014). This strategy also allows them to expand into other sectors of the black market.

For many years, the illicit drug market has been highly profitable and prominent. However, this oversaturation of the market forced some players to diversify into other illegal activity segments (Shelley, 2012). HT appeared as a logical area for expansion. The growing appeal of HT for OC networks can be understood from multiple viewpoints.

Some scholars once again turn to globalisation to explain the increased attractiveness of HT for OC networks. A side effect of globalisation is the widening of the wealth gap. In conjunction with advancing connectivity, people on the other side of the gap become more aware of the so-called "better life" and attempt to migrate by both legitimate and illegitimate means (Von Koppenfels, 2017). This creates more opportunities for OC networks to take advantage of and exploit people who see migration as their unique chance to escape. When it

comes to migration, the security imbalance contributes to the problem. Western democracies prioritize security to safeguard them from unmanaged migration flows, yet they fail to adequately protect migrants crossing borders (Sørensen, 2012). As a result, they become highly vulnerable to human trafficking.

Another angle to understand the appeal of HT to OC networks is through the viewpoint of Rational Choice Theory (RCT), which postulates that people decide whether to commit a crime by comparing the benefits and costs of engaging in crime (Cornish and Clarke, 2014). HT is a highly profitable crime. This is partly due to the "commodity" that HT provides. Unlike DT, it doesn't require constant replacement and allows for repeated exploitation (Shelley, 2012). Furthermore, HT is characterized as relatively low-risk due to the lack of proper regulations in many countries to contain crime and its high rewards. All this leads OC networks to opt for HT based on rational cost-benefit analysis (Wheaton, Schauer and Galli, 2010).

Overall, the two perspectives do not exclude each other. Moreover, to fundamentally understand the symbiosis between HT and DT, alluding to both perspectives is highly beneficial. The globalisation perspective explains the enhanced opportunities for HT, while the RCT perspective explains why OC networks take advantage of these opportunities. However, these explanations do not explain the more complex HT-DT symbiosis. They do not explain how one form of crime influences and amends another's functioning. This dissertation aims to fill this gap by elaborating on the use of drugs during HT.

## **2.2 Theoretical Background: Coercion and Methods of Control**

HT usually has a distinct multi-stage process (refer to UNODC definition on page 1). It begins with recruitment, followed by transportation, exploitation, and reintegration. As stated in the definition, coercion is a crucial part of HT and can occur in any of the first three stages. Control

and coercion methods exist in various forms, not just physical ones. Recent legal cases have demonstrated a variety of non-physical means of control, such as mental manipulation (Kim, 2007). Various theories and concepts can be alluded to explain the control element from different angles. In the context of intimate partner relations and domestic violence, the concept of coercive control emerged, signifying the intentional use of psychological, emotional, and financial abuse to maintain dominance (Barlow and Walklate, 2022). In the context of domestic abuse, it is performed by a man in relation to a woman, but mapping it to a HT, it is performed by the trafficker on a victim. The victim experiences patterns of systematic abuse, which act as a form of control and create invisible chains (Stark, 2024).

The Power and Control Wheel further builds on that knowledge. Similar to the coercive control concept, it was initially coined in the paradigm of domestic violence and put forward in 1984 by the Domestic Abuse Intervention Project. In 2010, the Wheel was adopted by the National Human Trafficking Resource Centre and Polaris to be representative of the control mechanisms used during HT (see *Figure 1*). Depending on the intended type of exploitation, the wheel illustrates the specific types of controls applicable to victims. According to the Wheel, induced drug addiction falls under the category of physical abuse. However, the original Power and Control Wheel, which was developed for domestic violence, has faced criticism for failing to account for the diverse socio-demographic contexts. This means that it is not as applicable as a concept to victims of minorities, as their experiences may be very different (Chavis and Hill, 2008).



Figure 4: Power and Control Wheel

Another interesting line of thought is to apply Biderman's framework of coercion to HT. The insight that it provides is the purpose of using particular methods of control (Baldwin, Fehrenbacher and Eisenman, 2014). Biderman first published his framework in 1957, outlining the psychological coercion methods used to manipulate the behaviour of war prisoners. These methods included isolation, monopolisation of perception, induced debility or exhaustion, threats, occasional indulgences, demonstration of omnipotence, degradation, and enforcing

trivial demands (Biderman, 1957). The DDD scheme, which stands for dependency, debility, and dread (Amnesty International Report on Torture, 1973), summarises these tactics of stress pattern manipulation. Debilitation refers to depriving the victim of food, social contact, and sleep. This leaves the victim highly vulnerable and dependent on the torturer. The dependency, in turn, induces a sense of dread. The dread comes in the form of the fear of the return of debilitation. Essentially, the entire process is based on “teaching” the victim new behaviours. The techniques induce a state of over-arousal and recondition the victim’s usual behaviour (*Amnesty International Report on Torture*, 1973). Some scholars noted the similarities between the techniques used during the interrogation of war prisoners and the tactics used to control HT victims.

A study on HT in Kentucky interviewed participants who had knowledge or experience with HT in Kentucky (Logan, 2007). This included law enforcement, health care providers, and service agencies. One of the study's contributions is to use Biderman's framework to improve HT understanding. The study linked the psychological coercion methods described in the framework to the experiences of HT victims. It also highlighted the purpose of using specific variations of coercion. Examples of achieving isolation include forbidding the victims from communicating and constantly keeping the victim in transit to avoid making any bonds. The trafficker aims to deprive the victim of any social support, thereby rendering them fully dependent on him. The trafficker achieves monopolisation of perspective by preventing the victim from learning the language of the country of exploitation and restricting access to cash or phones. This approach eliminates external stimuli and fosters introspection. Law enforcement, physical threats, and actual physical assaults are among the forms of threats used to cultivate anxiety and maintain control over victims. The use of occasional indulgences serves as positive reinforcement. They come in the form of promises to reduce the debt and promises to bring the family over. The demonstration of omnipotence implies the futility of resistance.

It can be achieved by having multiple abusers and reinforcing that nobody will believe the victim. Degradation reduces the victim's concerns to an 'animal level' and makes the cost of trying to escape seem excessively damaging. Numerous methods, such as cultural shame, sexual assault, and a lack of recognition for labour, can achieve this. Lastly, enforcing trivial demands, such as charging victims for basic necessities and enhancing contractual obligations, creates a habit of compliance. Traffickers deprive victims of necessities to weaken their mental and physical abilities, resulting in debility and exhaustion.

Another study denotes that debility can be induced by the unsolicited consumption of illicit substances (Baldwin, Fehrenbacher and Eisenman, 2014). However, it does not elaborate on the topic. The above-mentioned DDD scheme can be alluded to to better explain why control can be obtained by illicit substances. Cultivating a victim's substance addiction creates a dependency. Iteratively depriving and stimulating a victim with the drug induces a sense of debility and dread. Continuing the scheme reinforces the dependency. Despite the various theories and research papers written about the control element of human trafficking, none of them go into detail about the use of illicit substances. Some papers briefly mention this method but do not elaborate further. This dissertation aims to expand knowledge about the use of illicit substances as a means of control. It will do so by analysing trafficking patterns.

## **2.3 Human Trafficking: Geographical Patterns**

Scholars have dedicated extensive research to identifying HT patterns from various perspectives. Although this dissertation's focus is to identify patterns within the usage of illicit substances on HT victims, it could prove useful to tie in the findings with other patterns identified by scholars. For simplicity, from this point on, the HT patterns identified by other scholars will be referred to as the "existing patterns," and patterns within the usage of illicit substances on HT victims will be referred to as the "illicit substances patterns." Understanding

“existing patterns” is a useful base for interpreting “illicit substance patterns.” Furthermore, dissecting “illicit substance patterns” could offer a fresh perspective on the explanation of “existing patterns.”

By definition HT implies within-country or cross-border transfer of victims. Thus, HT has a very important spatial element to it. Understanding geographical patterns of HT can be highly beneficial to developing new strategies to counter the crime.

Although HT is getting more scholarly attention, it has been argued that for a long-time geography of HT in particular was an under researched area (Blazek, Esson and Smith, 2018). And while there exist multiple qualitative studies on the topic, quantitative studies are limited. This is partially due to the larger problem of the limited data available on HT, and even more limited availability of the geospatial data (Cockbain *et al.*, 2024).

A qualitative study on HT from West Africa to Europe provides some interesting insights. The study focuses on the cases of West African youth being tricked to come to Europe to play football professionally, which they see as an ultimate option to financially support their families (Esson, 2020). There are arguments made as to why this may or may not be considered HT. On the hand, youths agree to dislocate at their free will. On the other hand, it is argued that West African youth is targeted on the basis of fetishism and can be socially isolated and mistreated on arrival to the destination. This raises a broader issue of which of these cases are HT or not. There is a risk that due to the uncertainty certain HT cases are not recorded. Other studies looked at HT patterns from Mexico and Central America to the US for agricultural labour (Palacios and Yamamoto, 2017), and HT of Thai women to Singapore and Sydney (Yea, 2012).



# **METHODOLOGY**

## **Chapter III**

The present study aims to identify the victimisation patterns and profiles of victims controlled by psychoactive substances during HT. Due to the highly sensitive nature of HT, its privacy implications, and ethical considerations, it was not possible to collect primary data for this study. Thus, data was collected from secondary sources. The data was retrieved from a publicly available dataset. The dedicated section will delve deeper into the dataset's details. The present study divides its methodology into three sections. The first section will explore and present the data descriptively. Also, the limitations of the data and the respective mitigation strategies will be put forward. The second section will focus on the pre-processing stage and the treatment of limitations to minimize their impact on the analysis. The third section will describe the actual analysis steps.

### **3.1 Data**

The Counter Trafficking Data Collaborative (CTDC), the first global data hub on HT, provides the data used in the present study. Partners such as IOM UN Migration, Recollective, Polaris, Liberty Shared, A21, and the Observatory on Trafficking in Human Beings (OTSH) collect and provide the data to CTDC. For the sake of protecting the privacy of the victims, the data was put through a two-stage de-identification process. The first stage involved removing all identifying information through k-anonymisation. Microsoft Research created a privacy-preserving pipeline during the second stage. The product is a synthetic, fully anonymised dataset. The dataset consists of 193,025 HT cases. There are 37 variables that capture socio-

demographic information about the victims, the trafficking process, and the type of exploitation. Appendix I provides further information about each variable.

## **3.2 Data Limitations**

The dataset in its raw format had several limitations. The limitations were separated into three groups. The first group comprises the limitations that require acknowledgement. Simple pre-processing steps fixed the limitations in the second group. The limitations in the third group were problematic and required complex methodological mitigations.

### **3.2.1 Group One Limitations**

The "yearOfRegistration" variable documents the year in which one of the organisations providing the dataset assisted the HT victim. However, assistance does not always take place right after trafficking. Thus, this variable cannot be unquestionably used as an indicator of when the trafficking took place. The "gender" variable takes three values: female, male, and transgender/non-conforming. However, it's crucial to critically evaluate the data on transgender individuals. First, the inclusion of the transgender option in IOM's data dates back to 2017. Secondly, transgender people are chronically under-represented in such types of data. Lastly, the "CountryOfExploitation" variable only reflects the last country where the victim received assistance or identification. The variable only reflects the final country where the victim received assistance or identification.

### **3.2.2 Group Two Limitations**

The dataset is the product of collaboration between multiple large organisations. Each of the organisations has its own approach to collecting data. As a result, combining the data presents several challenges. The "ageBroad" variable denotes the victim's age during HT, while the

"traffickingMonths" variable represents the duration of exploitation. In the CTDC, these three variables are recorded at intervals. However, the CTDC presents the intervals on a non-equidistant scale. The reason for this is that all organisations recorded them according to their own decided intervals. This could potentially lead to problems during the analysis phase, necessitating the conversion of the variables to a different format.

### 3.2.3 Group Three Limitations:

Finally, the dataset's biggest limitation is the missing values. Pre-analysis data exploration showed that 84.13% of the data is missing. This arose because, in some cases, there was a lack of information about certain variables. This presented a significant issue that required attention. Several approaches to handling missing data were considered. To preserve the statistical significance of the dataset, Multiple Imputation by Chained Equations (MICE) was chosen as the best option. Multiple Imputation method takes the incomplete dataset and creates m-number of complete versions of the dataset by replacing missing values with plausible values (Van Buuren, 2012). The new datasets have the same given values, but different plausible values. The values will differ because of the uncertainty of the plausible values. Lastly, the results of the new datasets are pooled together (Van Buuren, 2012). In the best-case scenario, the pooled estimated are unbiased.

## 3.3 Pre-Processing

Data pre-processing was separated into three stages: 1) before MICE 2) MICE 3) after MICE.

### 3.3.1 Data Pre-processing Before MICE

As stated, before pre-processing, the dataset had 84.13% of the data missing. To conduct a comprehensive data analysis, it was crucial to investigate the missing data and comprehend the

underlying factors. Upon closer inspection, it was found that binary variables 8 through 37, which denote the trafficking process and exploitation information, take either a value of '1' or 'NULL', which corresponds to 'NA' in the dataset. A value of 1 indicates the existence of various forms of control and exploitation. 'NA' signifies that there is no information available regarding its presence or absence. As a result, binary variables do not evaluate to zero. This was problematic because MICE takes the values that are already present in the dataset and projects them on 'NAs'. Therefore, to apply MICE for this study, it was necessary to introduce zeros into binary variables 8 to 37.

The best solution for this problem was to identify cases where enough information existed to deduce where to put zeros. The binary variables 8 to 37 can be separated into more specific categories. Variables 8 through 22 represent control measures. Variables 23–25 indicated the purpose for which the victim was trafficked. Variable 23 indicates if the purpose was forced labour. Variable 24 indicates whether the purpose was sexual exploitation. Variable 25 indicates that the purpose was something else. If the victim was trafficked for forced labour and variable 23 takes the value of '1', then variables 26 to 30 can be referred to. They indicated the specific types of forced labour. If the victim was trafficked for sexual exploitation and variable 24 takes the value of '1', then variables 31 to 33 can be referred to. They indicate specific types of sexual exploitation. The last group of variables is 34–37. They describe the victim's relationship with their recruiter. This group of variables was decided to be removed from the dataset as they are not expected to have any relationship with the use of drugs on the victim.

To select the rows where 'NAs' will be replaced by zeros in columns with binary variables, the following conditional approach was adopted:

**Condition 1:** If a row has at least one '1' in columns 8 through 22

**Condition 2:** If column 23 takes the value of '1' and there is at least one '1' in columns 26 to 30

**Condition 3:** If column 24 takes the value of '1' and there is at least one '1' in columns 31 to 33

**Condition 4:** If columns 23 and 24 are 'NAs' and column 25 takes the value of '1'

If Condition 1 and any of Conditions 2 through 4 were met, then 'NAs' were replaced with zeros in that row. A function was built in R to identify and filter the rows. This worked because the rows included information on control mechanisms, trafficking purposes, and trafficking types.

The next step was to group countries into larger sub-regions in the "citizenship" and "CountryOfExploitation" variables. This was done for two reasons. Firstly, it was unlikely that there would be distinct trafficking patterns based on specific countries. Secondly, the MICE package in R allows a factored variable to only take on a limited number of values. The chosen regions were East Africa, East Asia and the Pacific, Eastern Europe and Central Asia, the Middle East and North Africa, North America, South America, South-West Asia, Southern Africa, West and Central Africa, and Western and Central Europe. Refer to Appendix 2 for information on which countries formed which groups.

### 3.3.2 Multiple Imputation by Chained Equations (MICE)

Due to the large number of missing values, a competent approach was required. Deletion and imputation are the two common approaches. Deletion was not an option for this study, as the data contained a large number of missing values and deleting them would result in a loss of

statistical significance (Cismondi *et al.*, 2013). Thus, Multiple Imputation was deemed an appropriate choice. There are a variety of imputation approaches, for example, replacing missing values with the mean or the median (Kaiser, 2014). However, a more advanced and precise approach was desired. Multiple Imputation is a powerful alternative approach to missing data. Studies found that even at a rate of 75% missingness, it outperforms deletion (Wulff and Jeppesen, 2017). The core principle behind Multiple Imputation is to predict missing values based on existing values (Stuart *et al.*, 2009). One popular method for Multiple Imputations is Multiple Imputations by Chained Equations (MICE). It was chosen in the present study because, compared to numerous other methods, it excels at handling large amounts of missing data.

The main analytical method of the present study is Unsupervised Machine Learning in a form of Principal Components Analysis, thus the purpose of the present study is explorative rather than inferential analysis, so the MICE process was simplified. The "mice" package in R was used (Van Buuren and Groothuis-Oudshoorn, 2011). To minimise the computational power required a minimum of imputed datasets and iterations was chosen. Three imputed datasets were created, each with five iterations of imputations. Imputation methods were chosen based on the class of each variable (see Table 1). After this, diagnostics were run on the imputed data. Convergence plots, where means and standard deviations for each variable are plotted against the number of iterations, are a commonly used MICE diagnostic. However, in this case convergence plots would not be very demonstrative as only five iterations were made (Van Buuren, 2012). Instead, kernel density estimates of the imputed and observed data were used (Van Buuren, 2012). Resulting kernel density plots showed that the distributions of observed and imputed data matched well (see Figure 2 for exemplary plots). The results were sufficient to continue the analysis. The results of the imputed datasets were then combined by calculating

the means for numeric values and majority voting was deployed for categorical data over the datasets.

| Variable class                 | Imputation methods             |
|--------------------------------|--------------------------------|
| Binary variables               | Logistic regression            |
| Numeric variable               | Bayesian linear regression     |
| Ordered categorical variables  | Proportional odds model        |
| Unordered categories variables | Polytomous logistic regression |

Table 1: Variable Classes and Corresponding Imputation Methods

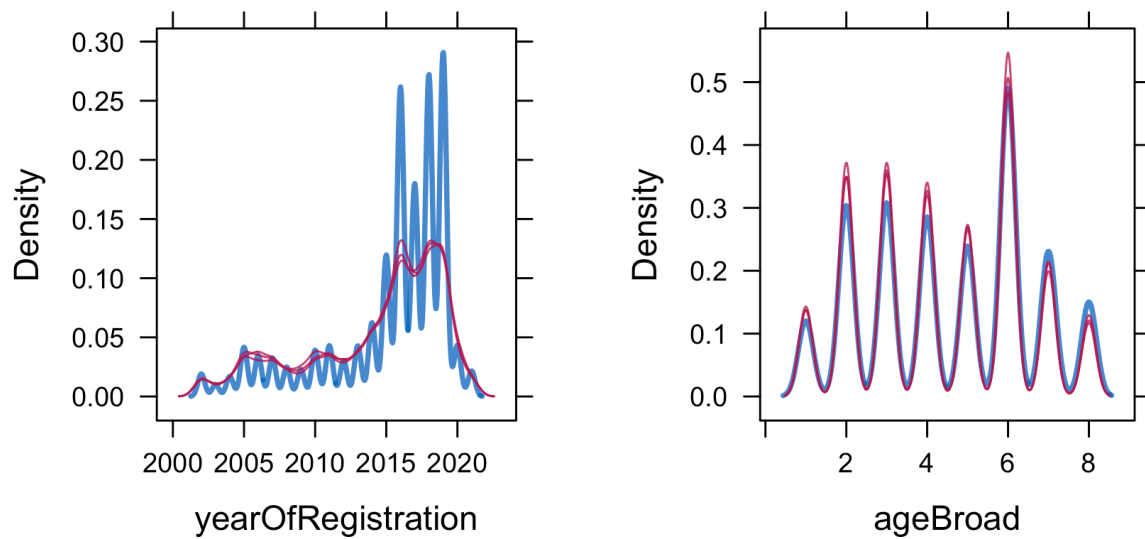


Figure 5: Kernel density estimates for the marginal distributions of the observed data (blue) and the  $m=3$  densities per variable calculated from the imputed data (thin red lines)

### 3.3.3 Data Post-processing After MICE

After acquiring the complete dataset, only cases where psychoactive substances were present were kept. This new dataset was modified to be suitable for clustering. All categorical values were converted to their corresponding levels and then changed to numeric values.

### **3.4 Explorative Data Analysis**

For data analysis, a random 10% sample was chosen from the data. This was done to reduce the computational power needed. To address the RQ and uncover the patterns of drug usage on the victims of HT, unsupervised machine learning was employed. The entirety of the analysis was done in RStudio. Several Machine Learning approaches were considered. Principal Components Analysis (PCA) was chosen as an ideal analysis method based on data and computational power limitations and complexity of the data.

PCA is a multivariate technique used to reduce the dimensionality of the data, while retaining the maximum fraction of the data's total variance (Groth *et al.*, 2012). It is particularly useful for multidimensional data visualization and exploration because it reduces the dimensionality of data to the most important components. The components are used to concisely visualise and explain the data (Groth *et al.*, 2012). The most important feature of PCA for this analysis is its ability to compute principal components loadings. Loadings show what features from the original dataset were most important in composing each of the principal components.

Other methods considered and attempted for the present study were K-means Clustering and Agglomerative Hierarchical Clustering. However, they were not chosen for a number of reasons. Even the 10% data sample was a rather large dataset with 32 variables in total. Deploying clustering and determining the ideal number of clusters through several Internal Validity Measures such as Silhouette method, Gap statistic, Calinski-Harabasz method



(Baarsch and Celebi, 2012) was too computationally intensive. Further decreasing the sample was also not an option, as it could lead to the loss of important information with the dataset.

## **RESULTS**

### **Chapter IV**

This section will focus on the results of PCA. It will look at the total variance explained by the first three principal components and elaborate on which factors are most influential in forming the principal components. An explanation of what makes each component unique and why it is the case will be provided. Furthermore, cluster profiles will be described.

#### **4.1 PCA: Explained Variance**

PCA was performed to understand the underlying structure of the data and to reduce the dimensionality. This process revealed some underlying patterns within the data. One of the biggest takeaways from the PCA is the estimation of the variance explained by each principal component. In total, the first three principal components only explain 35.08% of the variance cumulatively (see Figure 3), indicating that a significant amount of information is not captured by these components. It can be interpreted as a reflection of the data's complexity and the presence of diverse patterns spread across different dimensions, which was expected. This can be further linked to the complexity of HT as a crime and its lack of particular on-the-surface patterns.

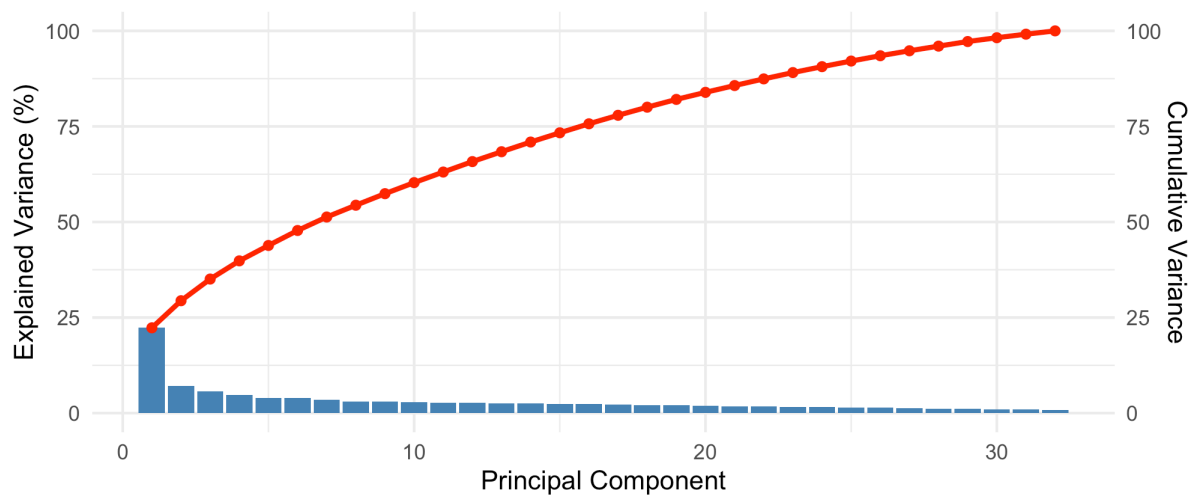


Figure 6: PCA Explained Variance, blue bars represent variance explained by each principal component, red line demonstrates cumulative variance

## 4.2 PCA: Principal Components Composition

### 4.2.1 Principal Component One: Conjunction with other types of control

The first principal component (PC1) explains 22.32% of the variance. To determine what original factors contributed most to its formation, the components' loadings were computed. The variables associated with other means of control have the highest positive loading scores: variable “meansOfControlPhysicalAbuse” has a loading of 0.29, variable “meansOfControlExcessiveWorkingHours” has a loading of 0.28, and variables “meansOfControlPsychologicalAbuse” and “meansOfControlDebtBondage” both have a loading of 0.27. This indicates that victims who scored higher on PC1 are not only controlled by psychoactive substances but tend to have multiple specific means of control used on them. The dataset contained 14 variables (besides illicit substances) representing unique means of control, however, the four means listed above had the highest loadings' magnitudes.

Firstly, certain classifications, such as the Power and Control Wheel, classify induced drug addiction as a form of control through physical abuse. Therefore, it's plausible that in certain

instances within the dataset, the presence of control via illicit substances also implies the presence of control through physical abuse.

Secondly, Biderman's framework of coercion (Biderman, 1957) provides an explanation for the presence of control through psychological abuse in cases with control through illicit substances. As stated in the Literature Review chapter, the central idea behind Biderman's framework of coercion is the dependency, debility, and dread scheme. If the perpetrator's goal is to make the victim dependent by inducing debility and subsequent dread, cultivating a drug addiction is one way to do so. However, the perpetrator can also use other methods, including isolation, monopolisation of perception, and the demonstration of omnipotence. They all fall under the category of control through psychological abuse. Perpetrators would likely employ multiple strategies to achieve the goal of the DDD scheme. This, in turn, would explain why, in cases where control is obtained through illicit substances, it is also obtained through psychological abuse.

The link between obtaining control through excessive working hours and obtaining control through illicit substances can also be explained logically. Different drugs have different effects. Some induce stimulation synthetically, if the goal of the perpetrator is to use victims for excessive physical labour, such as agriculture or construction, they can resort to stimulant drugs to enhance performance. Under the influence of drugs, victims will be more efficient and work for longer time intervals. This leads to the implementation of extended working hours.

Control through debt bondage can be a consequence of control through illicit substances.

After inducing a victim's addiction, perpetrators can demand that the victim start paying for illicit substances as they are in debt (Van Der Watt and Kruger, 2019).

#### 4.2.2 Principal Component Two: Age and Gender

The second principal component (PC2) explains a significantly smaller percentage of the variance (7.09%). The variables with the highest positive loading of 0.30 are “ageBroad” and “gender,” while “majorityStatusAtExploit” (indicates if the victim is a minor or an adult) has the highest negative loading of -0.30. They can all be grouped as demographic variables. The inverse proportionality of “ageBroad” and “majorityStatusAtExploit” is to be expected because “majorityStatusAtExploit” takes two values, “1” representing adults and “2” representing minors, while “ageBroad” records age in an ascending manner. In other words, victims with higher “ageBroad” scores have lower “majorityStatusAtExploit” scores, and vice versa.

Overall, cases with a higher PC2 score represent older men and cases with a lower PC2 score are younger women. One explanation for this distinction is that traffickers typically target men for physical labour, such as in agriculture and construction. It also makes sense to traffic older men rather than children. When it comes to women, they are more commonly trafficked for sexual exploitation, which explains why they tend to be younger.

#### 4.2.3 Principal Component Three: Geographical Distribution

The third principal component (PC3) explains 5.67% of the variance, variable “citizenship” has a loading of 0.46 and “CountryOfExploitation” has a loading of 0.44. This principal component is slightly more difficult to interpret because the “citizenship” and “CountryOfExploitation” variables are represented by broader regions and are recorded numerically. However, the numbers that represent each sub-region are entirely arbitrary. One way of interpreting PC3 is to hypothesise that victims of HT who score highly on PC3 are more

likely to have been trafficked within the sub-region of their citizenship. Consequently, victims who scored lower were trafficked to sub-regions where they do not hold citizenship.

### 4.3 PCA: Clusters

It was impossible to provide a visualisation of the data previously because of the number of dimensions that would be required. However, since PCA has reduced the dimensionality to the three principal components, it is possible to look at the 3D visualisation (see Figure 4). The 3D space reveals clusters that are not solidly distinct but visible. Each cluster is assigned a colour to make it easier to identify. Since a rotatable version of PCA cannot be provided, a table with the coordinates can be referred to (see Table 2 for approximate locations of clusters).

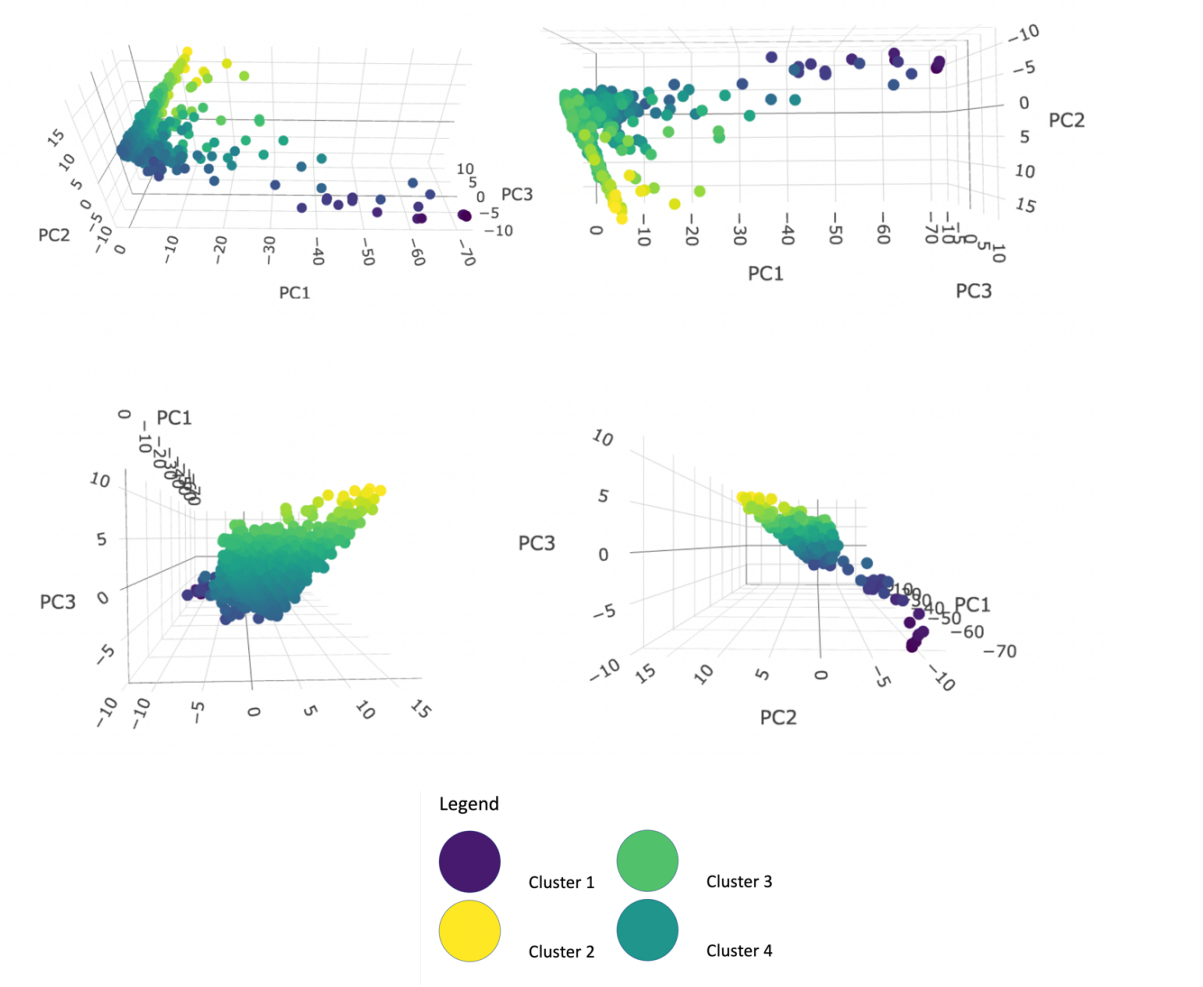


Figure 7: PCA 3D Visualisation (from different angles)

#### 4.3.1 Cluster One Description

The first cluster is relatively small, compared to others and is represented by the colour deep purple. It consists of cases where PC1 scores are lower than -35, which means that victims in this group are not controlled by other control methods besides illicit substances. This group has lower PC2 scores, which indicates that it mostly consists of younger women. The PC3 score for this group is also the lowest, which could potentially mean that they were trafficked outside the sub-region of their country of citizenship.

#### 4.3.2 Cluster Two Description

The second cluster is the second smallest one. It is represented by the colour yellow. It has a higher PC1 score, than the first cluster; however, it is still negative, meaning control methods besides psychoactive substances were not resorted to. All of the points in the yellow cluster have a positive PC2 score, which suggests that they are older men. The PC3 is positive for this cluster, meaning that victims were likely trafficked within the sub-region of their citizenship.

#### 4.3.3 Cluster Three Description

A light green colour represents the third cluster. The values for PC1 are neutral, as they take a value of zero, slightly below or above. This implies that other types of coercive control are only present in some cases. In PC2, the third cluster predominantly occupies positive values, indicating a high concentration of older men. The positive values for PC3 indicate that victim trafficking likely occurred within their citizenship sub-region.

#### 4.3.4 Cluster Four Description

The fourth cluster is represented by the colour dark green. In terms of PC1, it shares similar characteristics to the third cluster, also indicating the presence of other types of control. For

PC2, the fourth cluster is approximately situated between the values of -5 and 5. This means that there is no specification for age or gender for this group. The PC3 values are negative. Trafficking likely occurred outside the sub-region of the victims' citizenship.

| PCA Clusters            | Principal Components<br>Coordinates |           |           | Characteristics                                                                                                                                                                                                                                      |
|-------------------------|-------------------------------------|-----------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | PC1                                 | PC2       | PC3       |                                                                                                                                                                                                                                                      |
| Cluster 1<br>(purple)   | -35 to -70                          | -5 to -10 | -5 to -10 | <ol style="list-style-type: none"> <li>1. Not controlled by other control methods besides illicit substances</li> <li>2. Mostly consists of younger women</li> <li>3. Victims trafficked outside their country of citizenship</li> </ol>             |
| Cluster 2<br>(yellow)   | -10 to -20                          | 10 to 15  | 10 to 9   | <ol style="list-style-type: none"> <li>1. Rarely controlled by other control methods besides illicit substances</li> <li>2. Mostly consists of older men</li> <li>3. Likely trafficked within the sub-region of their citizenship</li> </ol>         |
| Cluster 3 (light green) | 0 to -15                            | 15 to -3  | 2 to 5    | <ol style="list-style-type: none"> <li>1. Other types of coercive control are present in some of the cases</li> <li>2. Mostly consists of older men</li> <li>3. Victims were likely trafficked within the sub-region of their citizenship</li> </ol> |

|                        |          |         |         |                                                                                                                                                                                                                                                             |
|------------------------|----------|---------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cluster 4 (dark green) | 0 to -15 | 5 to -5 | 1 to -5 | <ol style="list-style-type: none"> <li>1. Other types of coercive control are present in some of the cases</li> <li>2. No specification for age or gender</li> <li>3. Victims were likely trafficked outside the sub-region of their citizenship</li> </ol> |
|------------------------|----------|---------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Table 2: Results Summary

## DISCUSSION

### Chapter V

#### **Are there patterns within the usage of illicit substances as a means of control during human trafficking?**

Yes, but the identified patterns are still rather broad and further research and in-depth exploration would be highly beneficial.

Addressing the Hypotheses:

*Hypothesis<sub>1</sub>*: There are distinct geographical patterns of the usage of illicit substances as a means of control during human trafficking

PCA proved this hypothesis correct. Citizenship of the victim and the country of exploitation were significant factors in placing a victim in a specific group.

*Hypothesis<sub>2</sub>*: There are distinct demographic patterns of the usage of illicit substances as a means of control during human trafficking



PCA proved this hypothesis correct. The two most influential demographic factors in placing a victim in a specific group were age and gender.

*Hypothesis<sub>3</sub>*: There are distinct patterns of the usage of illicit substances as a means of control during human trafficking based on the type of exploitation the victim endures

PCA proved this hypothesis incorrect none of the type of labour variables were significant in forming principal components.

Lastly, a factor that was not anticipated to be a predictor of the usage of illicit substances on HT victims was the presence of other means of control. However, PCA showed that it was a significant factor in placing the victim in a certain group.

## **5.1 Key Findings**

PCA identified four cluster groups of HT cases in which the use of illicit substances served as a control strategy. Three principal components formed the clusters: 1) the presence of other means of control 2) victim demographic characteristics 3) geographical distribution. Based on the scores for the three principal components, profiles for each cluster were constructed. Although the nature of the patterns based on the above-mentioned characteristics is broad, it is possible to attempt to explain why the groups were formed the way they were.

The first group consists of younger women who were solely controlled by illicit substances and were trafficked outside the sub-region of their country of citizenship. The following interpretation can be applied to this group: Given the group's composition of young women, it is reasonable to assume that sexual exploitation was the primary reason for victimization. This conclusion is based on the fact that victims of HT sexual exploitation are predominantly female (De Cock and Woode, 2014). Other common purposes of exploitation for young women are

forced marriage and illicit adoption (Hume and Sidun, 2016). However, these types of HT rarely involve the use of drugs, thus are probably not represented in this group.

The second group is formed by older men, who were solely controlled by psychoactive substances and were trafficked within the sub-region of their citizenship. Because the group's composition is mostly older men, it can be assumed that they are exploited for physical labour such as agriculture or construction.

The third group is made up of older men, controlled by other means besides illicit substances, who were likely trafficked within the sub-region of their citizenship. This group is similar to the second group, and a similar conclusion can be drawn about victims being trafficked for intensive physical labour, such as construction and agriculture. However, the use of other control methods makes a distinction possible. It is possible that the exploitation of the second group takes place in more isolated areas, while the exploitation of the third group does not. The presence of multiple methods of control could indicate that employing one coercive technique is not enough to control the victims.

The fourth group does not have any tendencies in terms of gender or age. Victims of this group were controlled by other methods besides illicit substances and were likely trafficked outside the sub-region of their citizenship. Given the ambiguous data on gender and age, along with the existence of alternative forms of control like debt bondage, it's plausible that this group encompasses instances where victims engaged in diverse forms of labour, such as coerced criminal activities, thereby necessitating the use of various control methods.

## **5.2 Limitations**

Despite tailoring the study's methodology to tackle the challenges posed by the hidden nature of HT and limited data availability, it is crucial to acknowledge and reiterate the significant limitations.

#### 5.2.1 Inherent Data Limitations

Firstly, the highly sensitive nature of HT precluded the collection of primary data. Using secondary data restricted the scope of the present study to only exploring the variables in the CTDC dataset. Some of the variables created a challenge because of how they were recorded. For example, age was recorded in intervals instead of being a continuous variable. Furthermore, a large proportion of HT cases go unnoticed and unrecorded. Any data collected in regards to HT should be subject to critical examination, as it might not be fully representative of the issue's scope. Secondly, conducting a quantitative analysis of HT patterns was problematic due to limited data. The CTDC data used in the present study contained a large proportion of missing values. Although the dataset was imputed using the MICE algorithm, it needs to be emphasised that it is not a guarantee of perfect accuracy.

#### 5.2.2 Analysis Limitations

The analysis also had certain limitations. PCA only explained 35.08% of the variance. This means that there is still a large percentage of the variance that it did not account for. This is a serious limitation because if there was a possibility to conduct a more thorough analysis, potentially more specific patterns could have been uncovered. The next limitation closely relates to this one: the computational power required for a more intricate analysis. The capabilities of the computing system severely limited the present research.

### 5.3 Implications

The present study presents a solid foundation for further research on the patterns of usage of illicit substances by HT victims. The four identified groups of victims and the profiles of the groups can be used for further narrow research. Specifically, analysis of geographical patterns of HT within the sub-region of victims' citizenship and outside the subregion. Analysing not only the sub-regions but also specific countries within them can provide a more detailed account. One approach could be answering such questions as "How far within the subregion of the citizenship was the victim trafficked?" in each of the four identified groups. Further research can also focus on the reasons behind the identified patterns. Potentially exploring the different types of drugs used on victims in each of the four groups and understanding why the drugs were chosen.

## **CONCLUSION**

### **Chapter VI**

In conclusion, the present study has taken the pervasive problem of the interaction of HT and DT and identified an under researched area of the usage of illicit substances as a means of establishing control over the HT victims within it. The PCA proved to be a useful tool in visualising and explaining HT data effectively. It allowed for the identification of the three characteristic groups of factors: 1) Application of other means of control besides illicit substances 2) demographical age and gender factors 3) geographical distribution. The aim of this study has been accomplished and patterns were identified. However, results were limited by the restricted data availability. Due to this the results were more broad than desired. Yet they still encompassed the larger tendencies within the data. Having identified the central tendencies, it was possible to theorize on the reasons behind their formation. The direction for further researched have been identified which would allow for further elaborations on the usage of illicit substance as a means of control of HT victims.

## BIBLIOGRAPHY

1. *Amnesty International Report on Torture*. (1973). Available at:  
[http://books.google.ie/books?id=b4QxswEACAAJ&dq=amnesty+international+report+on+torture&hl=&cd=4&source=gsbs\\_api](http://books.google.ie/books?id=b4QxswEACAAJ&dq=amnesty+international+report+on+torture&hl=&cd=4&source=gsbs_api).
  
2. Armao, F. (2014). "Criminal Clusters: State and Organised Crime in a Globalised World," 1 (1), pp. 122–136. Available at:  
[https://iris.unito.it/bitstream/2318/148190/1/PROOF\\_Armao.pdf](https://iris.unito.it/bitstream/2318/148190/1/PROOF_Armao.pdf).
  
3. Aronowitz, A. A. (2009). *Human Trafficking, Human Misery*. Bloomsbury Publishing USA. Available at:  
[http://books.google.ie/books?id=3HbDEAAQBAJ&printsec=frontcover&dq=Human+trafficking,+human+misery:+the+global+trade+in+human+beings&hl=&cd=1&source=gsbs\\_api](http://books.google.ie/books?id=3HbDEAAQBAJ&printsec=frontcover&dq=Human+trafficking,+human+misery:+the+global+trade+in+human+beings&hl=&cd=1&source=gsbs_api).
  
4. Baarsch, J. and Celebi, M. E. (2012). "Investigation of Internal Validity Measures for K-Means Clustering." *Lecture Notes in Computer Science*, 2195 (1), pp. 471–476. Available at: [http://www.iaeng.org/publication/IMECS2012/IMECS2012\\_pp471-476.pdf](http://www.iaeng.org/publication/IMECS2012/IMECS2012_pp471-476.pdf).
  
5. Baldwin, S. B., Fehrenbacher, A. E. and Eisenman, D. P. (2014a). "Psychological Coercion in Human Trafficking." *Qualitative Health Research*, 25 (9), pp. 1171–1181. doi: 10.1177/1049732314557087.
  
6. Baldwin, S. B., Fehrenbacher, A. E. and Eisenman, D. P. (2014b). "Psychological Coercion in Human Trafficking." *Qualitative Health Research*, 25 (9), pp. 1171–1181. doi: 10.1177/1049732314557087.
  
7. Barlow, C. and Walklate, S. (2022). "What is 'coercive control'?" in *Routledge eBooks*, pp. 1–18. doi: 10.4324/9781003019114-1.
  
8. Biderman, A. D. (1957). "Communist attempts to elicit false confessions from Air Force prisoners of war." *PubMed*, 33 (9), pp. 616–25. Available at:  
<https://pubmed.ncbi.nlm.nih.gov/13460564>.

9. Black, J. (2011). *A Brief History of Slavery*. Hachette UK. Available at:  
[http://books.google.ie/books?id=NqyeBAAAQBAJ&printsec=frontcover&dq=A+Brief+History+of+Slavery:+A+New+Global+History&hl=&cd=1&source=gb\\_api](http://books.google.ie/books?id=NqyeBAAAQBAJ&printsec=frontcover&dq=A+Brief+History+of+Slavery:+A+New+Global+History&hl=&cd=1&source=gb_api).
10. Blazek, M., Esson, J. and Smith, D. P. (2018). "Relational geographies of human trafficking: Inequality, manoeuvring and im/mobility across space and time." *Transactions of the Institute of British Geographers*, 44 (1), pp. 63–78. doi: 10.1111/tran.12271.
11. Britain, G. (2015). *Modern Slavery Act 2015*. Available at:  
[http://books.google.ie/books?id=YLZ7zgEACAAJ&dq=Modern+Slavery+Act+2015&hl=&cd=1&source=gb\\_api](http://books.google.ie/books?id=YLZ7zgEACAAJ&dq=Modern+Slavery+Act+2015&hl=&cd=1&source=gb_api).
12. Chavis, A. Z. and Hill, M. S. (2008). "Integrating Multiple Intersecting Identities: A Multicultural Conceptualization of the Power and Control Wheel." *Women & Therapy*, 32 (1), pp. 121–149. doi: 10.1080/02703140802384552.
13. Christie, N. (1986). "The Ideal Victim." in *Palgrave Macmillan UK eBooks*, pp. 17–30. doi: 10.1007/978-1-349-08305-3\_2.
14. Cismondi, F., Fialho, A. S., Vieira, S. M., Reti, S. R., Sousa, J. M. C. and Finkelstein, S. N. (2013). "Missing data in medical databases: Impute, delete or classify?" *Artificial Intelligence in Medicine*, 58 (1), pp. 63–72. doi: 10.1016/j.artmed.2013.01.003.
15. Cockbain, E., Ashby, M., Bowers, K. and Zhang, S. X. (2024). "Concentrations of harm: Geographic and demographic patterning in human trafficking and related victimisation." *Criminology & Criminal Justice*. doi: 10.1177/17488958241245311.
16. Cornish, D. B. and Clarke, R. V. (2014). *The Reasoning Criminal: Rational Choice Perspectives on Offending*. Available at: <http://ci.nii.ac.jp/ncid/BA02924999>.
17. Daip. (2021). *Wheels - Domestic Abuse Intervention Programs. Domestic Abuse Intervention Programs*. Available at: <https://www.theduluthmodel.org/wheels/>.
18. De Cock, M. and Woode, M. (2014). "Profits and Poverty: the Economics of Forced Labour."

19. Decker, S. H. (2015). "Human trafficking: contexts and connections to conventional crime." *Journal of Crime and Justice*, 38 (3), pp. 291–296. doi: 10.1080/0735648x.2015.1039792.
20. *Drug trafficking*. (no date). *United Nations : Office on Drugs and Crime*. Available at: <https://www.unodc.org/unodc/en/drug-trafficking/index.html>.
21. Esson, J. (2020). "Playing the victim? Human trafficking, African youth, and geographies of structural inequality." *Population Space and Place*, 26 (6). doi: 10.1002/psp.2309.
22. Groth, D., Hartmann, S., Klie, S. and Selbig, J. (2012). "Principal Components Analysis." *Methods in Molecular Biology*, pp. 527–547. doi: 10.1007/978-1-62703-059-5\_22.
23. *Home - ECPAT*. (2023). *ECPAT*. Available at: <https://ecpat.org/>.
24. *Human Trafficking Power and Control Wheel*. (no date). *National Human Trafficking Hotline*. Available at: <https://humantraffickinghotline.org/en/resources/human-trafficking-power-and-control-wheel>.
25. *Human-Trafficking*. (no date a). *United Nations : Office on Drugs and Crime*. Available at: <https://www.unodc.org/unodc/en/human-Trafficking/Human-Trafficking.html>.
26. *Human-Trafficking*. (no date b). *United Nations : Office on Drugs and Crime*. Available at: <https://www.unodc.org/unodc/en/human-Trafficking/Human-Trafficking.html>.
27. Hume, D. L. and Sidun, N. M. (2016). "Human Trafficking of Women and Girls: Characteristics, Commonalities, and Complexities." *Women & Therapy*, 40 (1–2), pp. 7–11. doi: 10.1080/02703149.2016.1205904.
28. Kaiser, J. (2014). "Dealing with Missing Values in Data." *Journal of Systems Integration*, pp. 42–51. doi: 10.20470/jsi.v5i1.178.

29. Kim, K. (2007). "Psychological Coercion in the Context of Modern-Day Involuntary Labor: Revisiting *United States v. Kozminski* and Understanding Human Trafficking." *SSRN Electronic Journal*. Available at:  
[https://papers.ssrn.com/sol3/Delivery.cfm/SSRN\\_ID1021819\\_code332621.pdf?abstractid=1021819&mirid=1&type=2](https://papers.ssrn.com/sol3/Delivery.cfm/SSRN_ID1021819_code332621.pdf?abstractid=1021819&mirid=1&type=2).
30. Logan, Tk. (2007). *Human Trafficking in Kentucky*. Available at:  
[http://books.google.ie/books?id=91Z6nQAACAAJ&dq=Human+Trafficking+in+Kentucky&hl=&cd=1&source=gb\\_s\\_api](http://books.google.ie/books?id=91Z6nQAACAAJ&dq=Human+Trafficking+in+Kentucky&hl=&cd=1&source=gb_s_api).
31. O'Brien, E. (2013). "Ideal Victims in Trafficking Awareness Campaigns." in *Palgrave Macmillan UK eBooks*, pp. 315–326. doi: 10.1057/9781137008695\_21.
32. *Organized crime*. (no date). Available at:  
<https://www.interpol.int/en/Crimes/Organized-crime>.
33. Palacios, S. P. I. and Yamamoto, Y. (2017). "Trafficking in US Agriculture." *Antipode*, 49 (5), pp. 1306–1328. doi: 10.1111/anti.12330.
34. *Protocol to Prevent, Suppress and Punish Trafficking in Persons Especially Women and Children, Supplementing the United Nations Convention Against Transnational Organized Crime*. (2000). Available at:  
[http://books.google.ie/books?id=aCVBvgAACAAJ&dq=Protocol+to+Prevent,+Suppress+and+Punish+Trafficking+in+Persons&hl=&cd=1&source=gb\\_s\\_api](http://books.google.ie/books?id=aCVBvgAACAAJ&dq=Protocol+to+Prevent,+Suppress+and+Punish+Trafficking+in+Persons&hl=&cd=1&source=gb_s_api).
35. Rawley, J. A. and Behrendt, S. D. (2005). *The Transatlantic Slave Trade*. U of Nebraska Press. Available at:  
[http://books.google.ie/books?id=UQcOo4csdzEC&printsec=frontcover&dq=The+Transatlantic+Slave+Trade:+A+History&hl=&cd=1&source=gb\\_s\\_api](http://books.google.ie/books?id=UQcOo4csdzEC&printsec=frontcover&dq=The+Transatlantic+Slave+Trade:+A+History&hl=&cd=1&source=gb_s_api).
36. Schelling, T. C. (1984). *Choice and Consequence*. Harvard University Press. Available at:  
[http://books.google.ie/books?id=MF08nRe6jQoC&printsec=frontcover&dq=Choice+and+Consequence&hl=&cd=1&source=gb\\_s\\_api](http://books.google.ie/books?id=MF08nRe6jQoC&printsec=frontcover&dq=Choice+and+Consequence&hl=&cd=1&source=gb_s_api).
37. Serra, A. and Tagliaferri, R. (2019). "Unsupervised Learning: Clustering." in *Elsevier eBooks*, pp. 350–357. doi: 10.1016/b978-0-12-809633-8.20487-1.



38. Shelley, L. (2012). "The Relationship of Drug and Human Trafficking: A Global Perspective." *European Journal on Criminal Policy and Research*, 18 (3), pp. 241–253. doi: 10.1007/s10610-012-9175-1.
39. Sørensen, N. N. (2012). "Revisiting the Migration-Development Nexus: From Social Networks and Remittances to Markets for Migration Control." *International Migration*, 50 (3), pp. 61–76. doi: 10.1111/j.1468-2435.2012.00753.x.
40. Stark, E. (2024). "The Theory of Coercive Control." in *Oxford University Press eBooks*, pp. 248–282. doi: 10.1093/oso/9780197639986.003.0008.
41. Stuart, E. A., Azur, M., Frangakis, C. and Leaf, P. (2009). "Multiple Imputation With Large Data Sets: A Case Study of the Children's Mental Health Initiative." *American Journal of Epidemiology*, 169 (9), pp. 1133–1139. doi: 10.1093/aje/kwp026.
42. Sweileh, W. M. (2018). "Research trends on human trafficking: a bibliometric analysis using Scopus database." *Globalization and Health*, 14 (1). doi: 10.1186/s12992-018-0427-9.
43. *The Global Organized Crime Index 2021 | Global Initiative*. (2024). *Global Initiative*. Available at: <https://globalinitiative.net/analysis/ocindex-2021/>.
44. *The Global Organized Crime Index 2023 | Global Initiative*. (2024). *Global Initiative*. Available at: <https://globalinitiative.net/analysis/ocindex-2023/>.
45. UNODC. *Global Report on Trafficking in Persons 2022*. (2023). *Global report on trafficking in persons*. doi: 10.18356/9789210023351.
46. Van Buuren, S. (2012). *Flexible Imputation of Missing Data*. CRC Press. Available at: [http://books.google.ie/books?id=eIDNBQAAQBAJ&printsec=frontcover&dq=flexible+imputation+of+missing+data&hl=&cd=1&source=gbs\\_api](http://books.google.ie/books?id=eIDNBQAAQBAJ&printsec=frontcover&dq=flexible+imputation+of+missing+data&hl=&cd=1&source=gbs_api).
47. Van Buuren, S. and Groothuis-Oudshoorn, K. (2011). "mice: Multivariate Imputation by Chained Equations in R." *Journal of Statistical Software*, 45 (3). doi: 10.18637/jss.v045.i03.

48. Van Der Watt, M. and Kruger, B. (2019). "Breaking Bondages: Control Methods, 'Juju,' and Human Trafficking." in *Springer eBooks*, pp. 935–951. doi: 10.1007/978-3-319-63058-8\_54.
49. Vertovec, S. (2009). *Transnationalism*. Routledge eBooks. doi: 10.4324/9780203927083.
50. Von Koppenfels, A. K. (2017). "Human trafficking." in *Edward Elgar Publishing eBooks*. doi: 10.4337/9781783478651.00028.
51. Wheaton, E. M., Schauer, E. J. and Galli, T. V. (2010). "Economics of Human Trafficking." *International Migration*, 48 (4), pp. 114–141. doi: 10.1111/j.1468-2435.2009.00592.x.
52. Wilson, M. and O'Brien, E. (2016). "Constructing the ideal victim in the United States of America's annual trafficking in persons reports." *Crime Law and Social Change*, 65 (1–2), pp. 29–45. doi: 10.1007/s10611-015-9600-8.
53. Wulff, J. N. and Jeppesen, L. E. (2017). "Multiple imputation by chained equations in praxis: Guidelines and review." *The Electronic Journal of Business Research Methods*, 15 (1), pp. 41–56. Available at: [https://vbn.aau.dk/ws/files/257318283/ejbrm\\_volume15\\_issue1\\_article450.pdf](https://vbn.aau.dk/ws/files/257318283/ejbrm_volume15_issue1_article450.pdf).
54. Yea, S. (2012). "'Shades of grey': spaces in and beyond trafficking for Thai Women involved in commercial sexual labour in Sydney and Singapore." *Gender Place & Culture*, 19 (1), pp. 42–60. doi: 10.1080/0966369x.2011.617906.

## APPENDIX I: Variable Explanation Table

|    | Variable name           | Variable description                                                                                                                                                                                                                                                                | Variable type | Variable values                                                                          |
|----|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------------------------------------------------|
| 1. | yearOfRegistration      | The year in which the individual was registered and assisted by IOM or the year in which Polaris first received a report of the individual's trafficking situation, or the date of entry into Liberty Shared's VCMS.                                                                | numeric       | Range: [2002, 2021]                                                                      |
| 2. | gender                  | Designates the individual's expression or condition of being male, female, both or neither. This category considers the psychological, behavioral, social, and cultural aspects of being male, female, or non-binary (masculinity, femininity, neither, or a combination of both).  | string        | NULL, Male, Female                                                                       |
| 3. | ageBroad                | The individual's age at the time the individual was registered and assisted by IOM or Polaris first received a report of the individual's trafficking situation or at first point of contact with Liberty Shared's NGO partner.                                                     | string        | NULL<br><i>[missing data]</i> , 0—8, 9—17, 18—20, 21—23, 24—26, 27—29, 30—38, 39—47, 48+ |
| 4. | majorityStatusAtExploit | The individual's majority status when exploitation first began. Exploitation includes the exploitation of the prostitution of others or other forms of sexual exploitation, forced labour or services, slavery or practices similar to slavery, servitude or the removal of organs. | string        | NULL<br><i>[missing data]</i> , Minor<br><i>[Any person under the age of 18]</i> , Adult |

|    |                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                                                                                       |
|----|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------|
|    |                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        | [ <i>Any person who is 18 or older</i> ]                                              |
| 5. | citizenship           | The set of rights and duties that a person has with a country because of his/her legal bond with the country. This term is often used interchangeably with nationality. Liberty Shared records this as 'Country of Origin' as a proxy for citizenship. For cases with more than one citizenship, this dataset publishes the unique citizenship value based on the first reported field.                                                                                                        | string | Refer to a country list in Appendix 2                                                 |
| 6. | CountryOfExploitation | Indicates the country where a victim is first supported/assisted, identified and/or referred. In the context of human trafficking data, this is also referred to as the "country of destination2 (as opposed to "country of origin" if human trafficking was across borders). In the case of IOM, the last country of exploitation has a high number of missing values. Therefore, a proxy has been created to capture as much data as possible about the country the victim was exploited in. | string | Refer to a country list in Appendix 2                                                 |
| 7. | traffickMonths        | Indicates the reported duration of trafficking in months.                                                                                                                                                                                                                                                                                                                                                                                                                                      | string | NULL<br>[missing data], 0—12 (0-1 yr), 13—24 (1-2 yrs), 25—60 (2-5 yrs), 61+ (5+ yrs) |

|     |                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |                           |
|-----|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------|
|     |                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |                           |
| 8.  | meansOfControlDebtBondage        | Indicates whether the individual is forced to work to pay off a created or perceived debt. The individual is deceived to work for little or no pay, with no control over his/her debt. Debt bondage is defined as the status or condition arising from a pledge by a debtor of his personal services or of those of a person under his control as security for a debt, if the value of those services as reasonably assessed is not applied towards the liquidation of the debt or the length and nature of those services are not respectively limited and defined (United Nations' 1956 Supplementary Convention on the Abolition of Slavery). | binary<br>numeric | NULL<br>[missing data], 1 |
| 9.  | meansOfControlTakesEarnings      | Indicates whether the individual has experienced a situation where the exploiters have taken his/her remuneration in order to control him/her.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | binary<br>numeric | NULL<br>[missing data], 1 |
| 10. | meansOfControlThreats            | Indicates whether the individual experienced a situation in which his/her exploiter(s) explicitly or implicitly communicated an intent to inflict harm or loss on the individual or another. Threats to report or contact law enforcement in order to negatively impact the individual or another are considered distinct and are categorized in the <i>meansOfControlThreatOfLawEnforcement</i> variable.                                                                                                                                                                                                                                       | binary<br>numeric | NULL<br>[missing data], 1 |
| 11. | meansOfControlPsychologicalAbuse | Indicates whether the individual experienced a situation in which their exploiter(s) used emotionally abusive, deceptive, or devious tactics to influence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | binary<br>numeric | NULL<br>[missing data], 1 |

|     |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                           |
|-----|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------|
|     |                             | the individual. This may include, but is not limited to, name calling, verbal abuse, humiliating in front of others, manipulating real or perceived power imbalances, or shaming, blaming the individual. It may also include acts intended to exploit or prey upon any familial or romantic bonds/attachments the individual has with their exploiter(s).                                                                                                                                                                                                                                                                                           |                   |                           |
| 12. | meansOfControlPhysicalAbuse | Indicates whether the individual experienced a situation in which their exploiter(s) acted to cause physical injury, pain, disability, death or trauma to the individual. This includes but not limited to shoving, strangulation, shaking, slapping, punching, kicking, pulling hair, burning, branding or tattooing, the use of a weapon, or using one's size and strength against the individual. Physical acts of a sexual abuse are considered distinct and are categorized in the <i>meansOfControlSexualAbuse</i> variable.                                                                                                                   | binary<br>numeric | NULL<br>[missing data], 1 |
| 13. | meansOfControlSexualAbuse   | Indicates whether the individual experienced any kind of unwanted or non- consenting sexual contact from their exploiter(s) as a means by which to control the individual, rather than a purpose for which the individual was trafficked, in order to influence their behavior. This includes, but is not limited to, using sexual acts, assault, or contact as punishment or manipulation or normalizing sexual violence. It also includes coercive behavior that interferes with the individual's ability to control his/her reproductive life, including but not limited to, forcing/coercing the individual to terminate or continue a pregnancy | binary<br>numeric | NULL<br>[missing data], 1 |

|     |                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                   |                           |
|-----|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------|
|     |                                      | against their will, manipulating birth control, intentionally exposing someone to an STI, preventing condom negotiation, and/or attempting to or impregnating the individual without their consent.                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |                           |
| 14. | meansOfControlFalsePromises          | Indicates whether the individual was defrauded or tricked into entering the exploitative situation by their exploiter(s) using deception and false pretenses.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | binary<br>numeric | NULL<br>[missing data], 1 |
| 15. | meansOfControlPsychoactiveSubstances | Indicates whether the exploiter(s) induced the individual into substance abuse, provided substances to make the individual compliant or in order to influence their behavior, or exploited an existing substance abuse issue                                                                                                                                                                                                                                                                                                                                                                                                        | binary<br>numeric | NULL<br>[missing data], 1 |
| 16. | meansOfControlRestrictsMovement      | Indicates whether the exploiter(s) isolated, confined or limited the movement of the individual in any way physically or socially. This may include situations in which the individual is physically detained, prevented from moving without being accompanied or monitored, or the exploiter(s) threatens or enacts repercussions related to the individual's movement. This may also include forms of emotional isolation including restricting the individual's access to support systems or social networks or moving the individual frequently to prevent the individual from establishing support systems or social networks. | binary<br>numeric | NULL<br>[missing data], 1 |

|     |                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                           |
|-----|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------|
| 17. | meansOfControlRestrictsMedicalCare  | Indicates whether the exploiter(s) limited the individual's access to medical or health services. Including but not limited to necessary medical care or treatment being withheld, or when access to such treatment was controlled by the exploiter(s). This category also includes situations in which the individual was unable to access or interact with health services without being accompanied or monitored by the exploiter(s).                                                                             | binary<br>numeric | NULL<br>[missing data], 1 |
| 18. | meansOfControlExcessiveWorkingHours | Indicates whether the individual was required to work a significant number of hours more than what they were contracted or promised; could include over-time, late or atypical shifts, or overnight hours. Could be used a means for keeping the individual isolated and/or unable to seek help or report their situation. In some instances, work hours may be so excessive as to cause physical and/or mental health issues which may also limit the individual's capacity to seek help or report their situation. | binary<br>numeric | NULL<br>[missing data], 1 |
| 19. | meansOfControlThreatOfLawEnforce    | Indicates whether the individual experienced a situation in which their exploiter(s) explicitly or implicitly communicated an intent to contact or involve law enforcement or other relevant authorities, such as immigration authorities, in order to negatively impact the individual or another.                                                                                                                                                                                                                  | binary<br>numeric | NULL<br>[missing data], 1 |
| 20. | meansOfControlWithholdsNecessities  | Indicates whether the individual experienced a situation in which their exploiter(s) denied, restricted, or threatened to deny/restrict basic living necessities such as food, shelter, water,                                                                                                                                                                                                                                                                                                                       | binary<br>numeric | NULL<br>[missing data], 1 |



|     |                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                   |                           |
|-----|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------|
|     |                                  | hygiene, appropriate clothes, or necessary items for religious observance or gender expression.                                                                                                                                                                                                                                                                                                                                                                             |                   |                           |
| 21. | meansOfControlWithholdsDocuments | Indicates whether the individual experienced a situation in which their exploiter(s) limited, restricted, or controlled the individual's access to important documents including, but not limited to, the individual's passport, immigration documents, work permit, identification card, government benefit documents, birth certificate, gender identity carry letter, court-issued protection orders, custody papers, or other legal, official, or government documents. | binary<br>numeric | NULL<br>[missing data], 1 |
| 22. | meansOfControlOther              | Indicates whether the exploiter used a tactic intended to create or maintain power and control over the individual that cannot reasonably fit in any of the previous categories in <b>Variables 8-21</b> . Two variables in the CTDC k-anonymized dataset, "meansOfControlRestrictsFinancialAccess" and "meansOfControlUsesChildren", are grouped as other due to low response rates.                                                                                       | binary<br>numeric | NULL<br>[missing data], 1 |
| 23. | isForcedLabor                    | It indicates that the purpose for which a victim was trafficked was all work or service which was exacted from the individual, under the threat of a penalty and for which the individual has not offered himself or herself voluntarily. Sexual services are excluded from this definition.                                                                                                                                                                                | binary<br>numeric | NULL<br>[missing data], 1 |

|     |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                   |                           |
|-----|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------|
| 24. | isSexualExploitation    | It indicates that the purpose for which a victim was trafficked was sexual services, such as the exploitation of the prostitution of an individual. Typically, the exploiter(s) use force, fraud or coercion to achieve exploitation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | binary<br>numeric | NULL<br>[missing data], 1 |
| 25. | isOtherExploitation     | It indicates that the purpose for which a victim was trafficked was other type of exploitation, including those who underwent forced marriage, forced military, and organ removal. Typically, the exploiter(s) use force, fraud, or coercion to achieve exploitation. Polaris staff apply the U.S. federal definition of human trafficking as defined by the Trafficking Victims Protection Action (TVPA) to determine if a situation described through the helplines has indications of human trafficking. Situations of forced marriage, organ harvesting or “other”, which do not meet the U.S. definition of trafficking are not included in the data contributed by Polaris to the CTDC as Polaris currently does not collect data about these sub-types. | binary<br>numeric | NULL<br>[missing data], 1 |
| 26. | typeOfLabourAgriculture | Indicates whether the individual experienced forced labour in activities defined in ISIC Section A Division 01, “Crop and animal production, hunting and related service activities”. This category does not include work related to forestry and logging as defined by ISIC Section A, Division 02 which is considered distinct and categorized in <b>Variable 30</b> <i>typeOfLabourOther</i> .                                                                                                                                                                                                                                                                                                                                                              | binary<br>numeric | NULL<br>[missing data], 1 |

|     |                              |                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                           |
|-----|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------|
| 27. | typeOfLabour<br>Construction | Indicates whether the individual experienced forced labour in activities defined in ISIC Section F “Construction”.                                                                                                                                                                                                                                                                                                   | binary<br>numeric | NULL<br>[missing data], 1 |
| 28. | typeOfLabour<br>DomesticWork | Indicates whether the individual experienced forced labour in activities defined in ISIC Section T “Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use”.                                                                                                                                                                                     | binary<br>numeric | NULL<br>[missing data], 1 |
| 29. | typeOfLabour<br>Hospitality  | Indicates whether the individual experienced forced labour in activities defined in ISIC Section I “Accommodation and food service activities” including both Division 55 “Accommodation” and Division 56 “Food and beverage service activities”.                                                                                                                                                                    | binary<br>numeric | NULL<br>[missing data], 1 |
| 30. | typeOfLabour<br>Other        | Indicates whether the individual experienced forced labour of a specified type which could not reasonably be classified in any of the categories specified in <b>Variables 26-29</b> . Due to low response rates, the following variables in the CTDC k-anonymized data are grouped as other: aquafarming, begging, illicit activities, manufacturing, mining or drilling, peddling, and transportation and storage. | binary<br>numeric | NULL<br>[missing data], 1 |
| 31. | typeOfSexPros<br>titution    | Indicates whether the individual experienced exploitation in activities associated with an individual’s sexual act for payment.                                                                                                                                                                                                                                                                                      | binary<br>numeric | NULL<br>[missing data], 1 |

|     |                                  |                                                                                                                                                                                                                                                                                                                                                                                              |                   |                                   |
|-----|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------------------------|
| 32. | typeOfSexPornography             | Indicates whether the individual experienced exploitation in the production of visual material depicting sexual behavior that is intended to arouse sexual excitement in its audience and does not involve any participation from the audience. It does not include situations in which the audience remotely participates or interacts with the individual featured in the visual material. | binary<br>numeric | NULL<br><i>[missing data]</i> , 1 |
| 33. | typeOfSexOther                   | Indicates whether the individual experienced exploitation depicting sexual behavior other than prostitution and pornography. It includes remote interactive services and private sexual services in the CTDC k-anonymized data.                                                                                                                                                              | binary<br>numeric | NULL<br><i>[missing data]</i> , 1 |
| 34. | recruiterRelationIntimatePartner | Indicates whether a person who initially enticed or obtained the individual into the situation of exploitation was one with whom the individual has identified having a current or former romantic relationship.                                                                                                                                                                             | binary<br>numeric | NULL<br><i>[missing data]</i> , 1 |
| 35. | recruiterRelationFriend          | Indicates whether a person who initially enticed or obtained the individual into the situation of exploitation was one with whom the individual was familiar, exclusive of romantic partners, family relations, or other more formal relationships.                                                                                                                                          | binary<br>numeric | NULL<br><i>[missing data]</i> , 1 |
| 36. | recruiterRelationFamily          | Indicates whether a person who initially enticed or obtained the individual into the situation of exploitation was one with whom the individual was connected biologically, through marriage, or a                                                                                                                                                                                           | binary<br>numeric | NULL<br><i>[missing data]</i> , 1 |

|     |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |                           |
|-----|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------|
|     |                        | person who the individual has identified as having been their current or former custodian or guardian. This may include, but is not limited to, parents, primary caregivers, and foster parents.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                |                           |
| 37. | recruiterRelationOther | Indicates whether a person who initially enticed or obtained the individual into the situation of exploitation was a person with whom the individual had any other notable relationship that cannot reasonably fit into previous categories. This may include, but is not limited to, labor brokers, contractors, formal employers, or smugglers. In the case of Liberty Shared, data collected on recruiters was more extensive than Global Dataset required. Liberty Shared transformed values to match IOM values, resulting in a higher level of 'other' values where information recorded did not match (for example – Broker – becomes other). | binary numeric | NULL<br>[missing data], 1 |

Source: Counter Trafficking Data Collaborative (CTDC) Synthetic Data Codebook. Version 1 – September 2021

## **APPENDIX II: Countries Grouped by Sub-Regions** (only contains countries that are present in the CTDC dataset)

| <b>Broader Region</b> | <b>Countries</b>                                                                                                                                                                                 |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| East Africa           | Burundi; Comoros; Djibouti; Eritrea; Ethiopia; Kenya; Madagascar; Malawi; Mauritius; Mozambique; Rwanda; Seychelles; Somalia; South Sudan; Uganda; United Republic of Tanzania; Zambia; Zimbabwe |

|                                 |                                                                                                                                                                                                                                                    |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| East Asia and Pacific           | Australia; Brunei Darussalam; Cambodia; China; Hong Kong; Indonesia; Japan; Korea; Lao People's Democratic Republic; Malaysia; Mongolia; Myanmar; Philippines; Singapore; Taiwan; Thailand; Viet Nam                                               |
| Eastern Europe and Central Asia | Armenia; Azerbaijan; Belarus; Estonia; Georgia; Hungary; Kazakhstan; Kyrgyzstan; Lithuania; Latvia; Moldova; Poland; Russian Federation; Slovakia; Slovenia; Tajikistan; Turkmenistan; Ukraine; Uzbekistan                                         |
| Middle East and North Africa    | Algeria; Bahrain; Egypt; Iran (Islamic Republic of); Iraq; Israel; Jordan; Kuwait; Lebanon; Libya; Morocco; Oman; Qatar; Saudi Arabia; Syrian Arab Republic; Tunisia; United Arab Emirates; Yemen                                                  |
| North America                   | Canada; Mexico; United States                                                                                                                                                                                                                      |
| South America                   | Argentina; Bolivia (Plurinational State of); Brazil; Chile; Colombia; Ecuador; Guyana; Peru; Paraguay; Suriname; Uruguay; Venezuela (Bolivarian Republic of)                                                                                       |
| South-West Asia                 | Afghanistan; Bangladesh; Bhutan; India; Maldives; Nepal; Pakistan; Sri Lanka                                                                                                                                                                       |
| Southern Africa                 | Botswana; Eswatini; Lesotho; Namibia; South Africa                                                                                                                                                                                                 |
| West and Central Africa         | Angola; Benin; Burkina Faso; Cameroon; Central African Republic; Chad; Congo; Côte d'Ivoire; Democratic Republic of the Congo; Gabon; Gambia; Ghana; Guinea-Bissau; Guinea; Liberia; Mali; Mauritania; Niger; Nigeria; Senegal; Sierra Leone; Togo |
| Western and Central Europe      | Austria; Belgium; Czechia; Denmark; Finland; France; Germany; Greece; Iceland; Ireland; Italy; Luxembourg; Netherlands (Kingdom of the); Norway; Portugal; Spain; Sweden; Switzerland; United Kingdom.                                             |

Source: UNICEF Regional Classification<sup>1</sup>

<sup>1</sup> <https://data.unicef.org/regionalclassifications/>

